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Editors
SIAM Journal on Matrix Analysis and Applications

Re: Submission of "A Channel-Conditioned Perturbation Theorem for Hidden Schur-Complement Resolvents"

Dear Editors,

I am pleased to submit the manuscript "A Channel-Conditioned Perturbation Theorem for Hidden Schur-Complement Resolvents" for consideration by SIAM Journal on Matrix Analysis and Applications.

The manuscript develops a restricted inverse-channel perturbation theorem for hidden Schur-complement resolvents. Its main contribution is a channel-conditioned bound that combines inverse amplification and singular-subspace separation, together with sharpness, normal-form, independent-variation, and composite-map impossibility results.

The paper is intended for SIMAX because its core objects are singular subspaces, Wedin-type separation, resolvent perturbation, matrix/operator perturbation bounds, and finite-dimensional numerical verification. The control and model-reduction consequences are presented as applications of the matrix-analysis theory.

To the best of my knowledge, the manuscript is original, is not under consideration elsewhere, and all computational tables are generated by the included appendix script. A public preprint or working-paper version may be posted according to journal policy and disclosed as appropriate.

Thank you for considering the manuscript.

Sincerely,

Philip M. Parker